DiD and Synthetic Controls

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1 Recap

Treatment allocation cannot be randomized, time series are available:

- DiD
- Synthetic control + advancements
- Regression discontinuity design (RDD)

1.1 Brief recap of the main idea

- Comparing outcomes for treated and control *after* intervention

$$\mathbb{E}[Y_{igt}|t = \text{post}, g = 1] - \mathbb{E}[Y_{igt}|t = \text{post}, g = 0] = \beta + (\alpha_{g=1} - \alpha_{g=0}) + (\delta_{\text{post}} - \delta_{\text{post}})$$

- Comparing outcomes for treated *before* and *after* intervention

$$\mathbb{E}[Y_{igt}|t = \text{post}, g = 1] - \mathbb{E}[Y_{igt}|t = \text{pre}, g = 1] = \beta + (\delta_{\text{post}} - \delta_{\text{pre}}) + (\alpha_{g=1} - \alpha_{g=1})$$

- Difference-in-difference estimator

$$\text{ATET}_{\text{DiD}} = \beta$$

1.1.1 Common or parallel trend assumption

- For the simple 2 groups - 2 time points setting the assumption is formally given by

$$\begin{aligned}\delta_{\text{post}} - \delta_{\text{pre}} &= \mathbb{E}[Y_{igt}(0)|t = \text{post}, g = 1] - \mathbb{E}[Y_{igt}(0)|t = \text{pre}, g = 1] \\ &= \mathbb{E}[Y_{igt}(0)|t = \text{post}, g = 0] - \mathbb{E}[Y_{igt}(0)|t = \text{pre}, g = 0]\end{aligned}$$

  - The difference between pre and post (potential) outcomes is equal between treatment and control group (and under the typical assumptions of SUTVA this relates to observed outcomes)
- What happens when the assumption fails?

$$\text{ATET}_{\text{DiD}} = \beta + (\text{difference in trends})$$

1.2 Other assumtions / limitations of DiD

- No other (unobserved) events happened close to treatmend and could cause a change in the post treatment period (i.e., that differentially impact treated and control units)
- There are no compositional changes over time in the groups
- Full compliance for the ATET: all treatment units actually receive treatment (otherwise DiD is an ITT)
- No anticipation of the treatment and self-selection into groups based on treatment
- The standard DiD does not allow heterogeneous treatment effects (when a treatment impacts individuals differently)
  - Causal effects are assumed constant across individuals and time
- Not all units may receive treatment at the same time (differential timing, staggered designs)
  - More advanced methods necessary

1.2.1 Additional notes on DiD

Basic DiD estimates an average level shift, not a trend change

- The estimated treatment effect is an average post-treatment effect on treated units
- It averages over the whole post period
- It basically assumes a jump and then continuance of the original trend

Potential alternatives:

- Estimate a DiD for each post-treatment period
- Linear trend-break model

2 Synthetic Controls

2.1 Key idea

- Again: compare the time series of the outcomes of the treated to a control (group), but
  - We have only one (or a few) treated unit(s)
  - We have several (many) control units
  - All individually or the average are not comparable to the treated unit (e.g., no common trends)
    - We cannot use DiD estimation or matching
- Proposition: When the units of analysis are a few treated units, a combination of comparison units (a "synthetic control") often does a better job reproducing the characteristics of a treated unit than any single comparison unit alone
  - Instead of finding one matching parner, we construct a matching partner out of many controls
- A combination of DiD, matching and weighting
  - Instead of (just) matching on covariates, we try to create a matching pre-treatment trend by weighting several control units

The setting:

- Suppose that we observe a panel of $i \in \{0, 1, 2, \ldots, J\}$ units in periods $t \in \{1, 2, 3, \ldots, T\}$
- Unit "zero" is "treated" and the remining J units are an untreated reservoir of potential controls
- We have again potential outcomes $Y_{it}(1)$ and $Y_{it}(0)$ and corresponding observed outcomes for treated and controls Y_{it}
- Moreover, we have a vector of k pre-intervention covariates or control variables $X_i(k \times J + 1)$
- Finally, treatment happens at time t^* distinguishing periods before (T_0) and after (T_1) treatment

Very similar to the DiD, but ...

- We only look at one treated unit
- We do not assume parallel trends
- We need to observe some characteristics of X_i of the treatment and control units

- We try to create an artificial (synthetic) control unit that matches the trend of the treatment unit in T_0 (i.e., pre-treatment)
  - This comparison unit is created as a weighted average of all potential comparison units that best resembles the characteristics of the treated unit(s)
- Let $W = (\omega_1, \ldots, \omega_J)'$
  - Each W represents a potential synthetic control
  - All weights in W need to be positive $\omega_j > 0$
  - Weights in W sum up to one: $\sum_{i=0}^{J} \omega_i = 1$
- The vector $W = (\omega_1, \ldots, \omega_J)'$ is chosen to minimize $||X_0 - X_1 W||$,
  - Subject to the above weight constraints
  - With X_1 being the X_i for all J units that are untreated (in the pre-treatment period T_0)
- Given the estimated weights W^* we can extrapolate the matched trend to the post-treatment period

2.2 Synthetic control – the treatment effect

- The treatment effect will be an ATET (average effect on the treated)
- For an after-treatment period t (with $t > t^*$) the synthetic control estimator is

$$\hat{\tau}_{0t} = Y_{0t} - \sum_{j=1}^{J} \omega_j^* Y_{jt}$$

  - i.e., the difference in observed outcomes for treated and untreated units, whereby all the untreated units are weighted with the estimated weights w_j^*
- And more commonly we estimate and average post-treatment effect as

$$\hat{\tau}^{\text{SC}} = \frac{1}{T - T_0} \sum_{t > T_0} \hat{\tau}_{0t}$$

2.3 How to estimate the weights

- A nested optimization problem using actually two sets of weights W and Y
- We minimize the norm $||X_0 - X_1 W||$ as

$$\arg\min_{w} ||X_0 - X_1 W|| = \sqrt{\sum_{m=1}^{k} v_m \left(X_{1m} - \sum_{j=1}^{J} \omega_j X_{jm} \right)^2}$$

or

$$(X_0 - X_1 W)' V (X_0 - X_1 W)$$

  - The initial weight constraints (i.e., $\omega_j \geq 0$ and $\sum_{i=1}^{J} \omega_i = 1$) are important as they help with regularization and avoid some overfitting
- The second set of weights V represents the covariance importance
  - The v_m are covariate-specific weights that reflect the relative importance of the variable when we measure the difference between the treated unit and the synthetic control trends
  - They provide a link to the actual outcome that we want to predict (we do not only want a match between covariates but pre-treatment outcomes)

The idea of the synthetic control is to reproduce the behavior of the outcome variable of the treated unit (and not only their characteristics for X_1)

- Reflect the predictive importance of the covariates for the outcome
- Usually, they are chosen to minimize the mean squared prediction error on the pre-treatment outcomes Y

$$\sum_{t=1}^{T_0} \left(Y_{0t} - \sum_{j=1}^{J} \omega_j^*(V) Y_{jt} \right)^2$$

- Commonly, they are also chosen via out-of-sample validation to avoid overfitting
- It is required that all $v_m > 0$

2.4 Example – Cigarette consumption

Estimating the effects of California's proposition 99

- In 1988 a voter initiative in California known as proposition 99, initiated the first modern-time large-scale tobacco control program in the US
- It increased California's cigarette excise tax by 25 cents per pack
- The tax revenues went to health and anti-smoking education budgets, funding anti-smoking media campaigns, and spurred local clean indoor-air ordinances throughout the state

Was there an effect of this law on cigarette consumption?

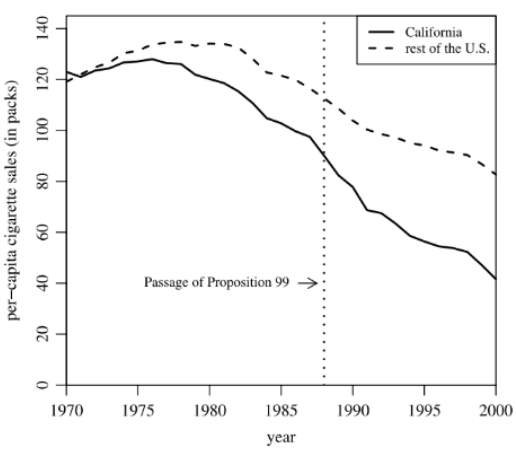
- Outcome variable of interest
  - Annual per capita cigarette consumption at the state level

Data

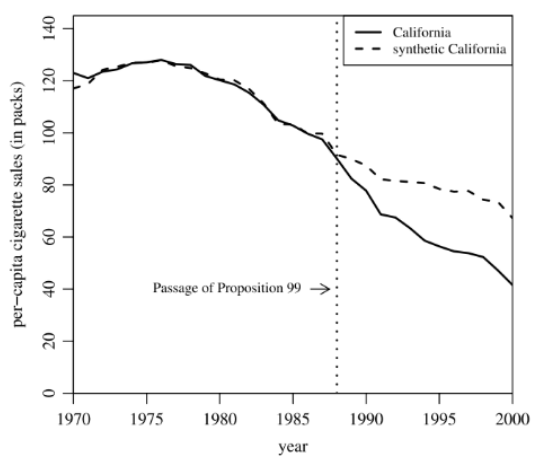
- Annual state-level panel data for the period 1970-2000
- Proposition 99 was passed in November 1988 and went into effect in January 1989
  - 19 years of pre-treatment data T_0
- Data frame was chosen to end in year 2000, because at about this time anti-tobacco measures were implemented across many states in the US, invalidating them as potential control units
- 39 states (i.e., California and a donor pool of 38 states)

Recall that the synthetic California should be constructed as a weighted average of potential control states

- Weights are chosen to that the "synthetic California" best reproduces the values of a set of predictors of cigarette consumption in California before the passage of proposition 99
- Because the "synthetic California" is meant to reproduce the smoking rates that would have been observed for California in the absence of proposition 99 (i.e., the counterfactual $Y(0)$)
  - Authors discard from the donor pool states that adopted some other large-scale tobacco control program or tax increases during the sample period
- Control units need to be free of any similar kind of treatment during the pre and post treatment periods



California vs. the control states

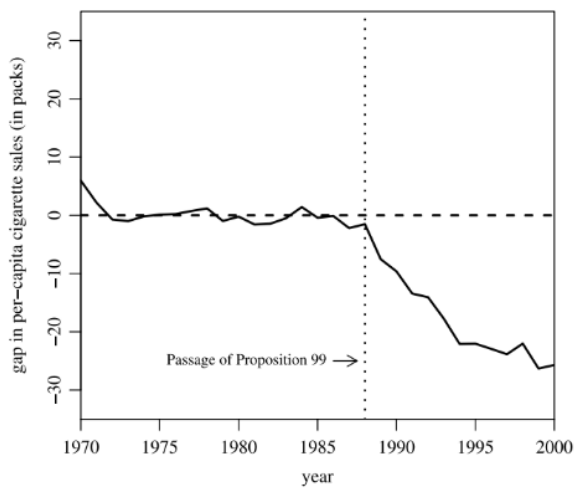


California vs. synthetic control

2.4.1 Difference (gap) in per-capita cigarette sales

Only 5 states made up the final synthetic control

- California prior to the treatment is best reproduced by a combination of Colorado, Connecticut, Montana, Nevada and Utah
- All other states in the donor pool are assigned zero W-weights



California vs. the control states

2.5 Inference

Traditionally, a form of permutation approach is performed

- Estimte the treatment effect using the same synthetic control procedure using the other control units as treated (i.e., placebo tests)
  - If the focal treatment units' effects are unusually large relative to the gaps for the other control units that did not implement treatment, we have confidence in our effect
  - However, if other control units can produce similarly large effects, our focal treatment effect might just be caused by random chance